Lecture 16: Section 5.2

Antiderivatives

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What is an Antiderivative?

Definition (Antiderivative)

A function F is called an **antiderivative** of f on an interval I if

$$F'(x) = f(x)$$

for every x in I.

Key Idea

- Antidifferentiation is the **reverse process** of differentiation
- If $F'(x) = f(x)$, then F is an antiderivative of f
- Also called "finding the primitive" or "integration"

Example

- $F(x) = x^2$ is an antiderivative of $f(x) = 2x$ because $\frac{d}{dx}(x^2) = 2x$
- $F(x) = \sin x$ is an antiderivative of $f(x) = \cos x$ because $\frac{d}{dx}(\sin x) = \cos x$

Family of Antiderivatives

Theorem (Constant of Integration)

If F is an antiderivative of f on an interval I, then the **most general antiderivative** *of f on I is*

$$F(x) + C$$

where C is an arbitrary constant.

Why the $+C$?

- Derivative of a constant is zero: $\frac{d}{dx}(C) = 0$
- If $F'(x) = f(x)$, then $\frac{d}{dx}(F(x) + C) = f(x)$ for any constant C
- All antiderivatives differ by a constant

Example

All antiderivatives of $f(x) = 2x$ are of the form $F(x) = x^2 + C$

Example 1: Finding Antiderivatives

Find all antiderivatives of $f(x) = 3x^2$.

Solution:

1. We need $F(x)$ such that $F'(x) = 3x^2$
2. Recall: $\frac{d}{dx}(x^3) = 3x^2$
3. So $F(x) = x^3$ is one antiderivative
4. Most general: $F(x) = x^3 + C$

Verification:

$$\frac{d}{dx}(x^3 + C) = 3x^2 + 0 = 3x^2$$

Answer: All antiderivatives are $x^3 + C$, where C is any constant.

Indefinite Integral Notation

Definition (Indefinite Integral)

The **indefinite integral** of f with respect to x is denoted by

$$\int f(x)\,dx$$

and represents the family of all antiderivatives of f.

Notation Explained

- $\int$ is the **integral sign**
- $f(x)$ is the **integrand**
- dx indicates integration with respect to x
- The result is $F(x) + C$, where $F'(x) = f(x)$

$$\int 2x\,dx = x^2 + C \quad \text{because} \quad \frac{d}{dx}(x^2 + C) = 2x$$

Basic Integration Formula

For any real number $n \neq -1$:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

Check by differentiation:

$$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1} + C\right) = \frac{(n+1)x^n}{n+1} = x^n$$

- $\int x^2 \, dx = \frac{x^3}{3} + C$
- $\int x \, dx = \frac{x^2}{2} + C$
- $\int 1 \, dx = \int x^0 \, dx = x + C$

Example 2: Basic Indefinite Integrals

Evaluate each indefinite integral:

a) $\int x^4\,dx$

b) $\int \sqrt{x}\,dx$

c) $\int \frac{1}{x^2}\,dx$

Solution:

a) $\int x^4\,dx = \frac{x^{4+1}}{4+1} + C = \frac{x^5}{5} + C$

b) $\int \sqrt{x}\,dx = \int x^{1/2}\,dx = \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C$

c) $\int \frac{1}{x^2}\,dx = \int x^{-2}\,dx = \frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$

Verification:

- $\frac{d}{dx}\left(\frac{x^5}{5}\right) = x^4$
- $\frac{d}{dx}\left(\frac{2}{3}x^{3/2}\right) = x^{1/2} = \sqrt{x}$
- $\frac{d}{dx}\left(-\frac{1}{x}\right) = \frac{1}{x^2}$

Basic Integration Formulas

Essential Formulas

$$\int k\,dx = kx + C \quad \text{(Constant)}$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \quad \text{(Power Rule)}$$

$$\int e^x\,dx = e^x + C \quad \text{(Exponential)}$$

$$\int \frac{1}{x}\,dx = \ln|x| + C \quad \text{(Special Case)}$$

Trigonometric Integration Formulas

Basic Trigonometric Integrals

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

Example 3: Using Integration Formulas

Evaluate:

a) $\int (3x^2 + 4x + 1)\, dx$

b) $\int (\cos x - 2\sin x)\, dx$

c) $\int \left(e^x + \frac{1}{x}\right)\, dx$

Solution:

a) $\int (3x^2 + 4x + 1)\, dx = 3 \cdot \frac{x^3}{3} + 4 \cdot \frac{x^2}{2} + x + C = x^3 + 2x^2 + x + C$

b) $\int (\cos x - 2\sin x)\, dx = \sin x - 2(-\cos x) + C = \sin x + 2\cos x + C$

c) $\int \left(e^x + \frac{1}{x}\right)\, dx = e^x + \ln|x| + C$

Verification: Differentiate each result to check!

Algebraic Properties of Integrals

Theorem (Linearity Properties)

Let f and g be functions and k be a constant. Then:

1. $\int k \cdot f(x)\,dx = k \int f(x)\,dx$ *(Constant Multiple)*
2. $\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$ *(Sum Rule)*
3. $\int [f(x) - g(x)]\,dx = \int f(x)\,dx - \int g(x)\,dx$ *(Difference Rule)*

Important Notes

- There is **no product rule** or **quotient rule** for integration!
- These properties allow us to integrate term-by-term
- The constant C appears only once at the end

Example 4: Using Properties

Evaluate $\int (4x^3 - 3x^2 + 2x - 5)\,dx$

Solution:

$$\begin{aligned}\int (4x^3 - 3x^2 + 2x - 5)\,dx &= \int 4x^3\,dx - \int 3x^2\,dx + \int 2x\,dx - \int 5\,dx \\ &= 4\int x^3\,dx - 3\int x^2\,dx + 2\int x\,dx - 5\int 1\,dx \\ &= 4\cdot\frac{x^4}{4} - 3\cdot\frac{x^3}{3} + 2\cdot\frac{x^2}{2} - 5x + C \\ &= x^4 - x^3 + x^2 - 5x + C\end{aligned}$$

Verification:

$$\frac{d}{dx}(x^4 - x^3 + x^2 - 5x + C) = 4x^3 - 3x^2 + 2x - 5$$

Example 5: More Complex Integration

Evaluate $\int \left(3e^x + 2\cos x - \frac{4}{x^2}\right)\, dx$

Solution:

$$\int \left(3e^x + 2\cos x - \frac{4}{x^2}\right)\, dx = 3\int e^x\, dx + 2\int \cos x\, dx - 4\int x^{-2}\, dx$$

$$= 3e^x + 2\sin x - 4\cdot\frac{x^{-1}}{-1} + C$$

$$= 3e^x + 2\sin x + \frac{4}{x} + C$$

Verification:

$$\frac{d}{dx}\left(3e^x + 2\sin x + \frac{4}{x} + C\right) = 3e^x + 2\cos x - \frac{4}{x^2}$$

What are Integral Curves?

Definition (Integral Curves)

The graphs of antiderivatives of a function f are called **integral curves** of f.

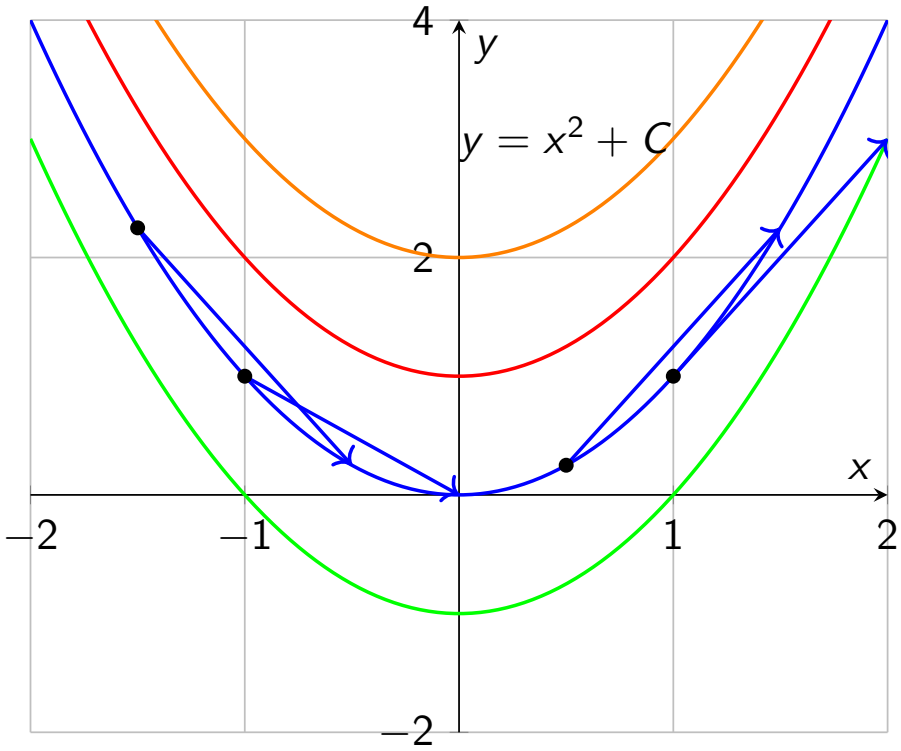
Properties of Integral Curves

- All integral curves of f are **vertical translations** of each other
- They all have the **same slope** at each x-value
- The slope at point (x, y) on any integral curve is $f(x)$
- The family is given by $y = F(x) + C$

Example

All integral curves of $f(x) = 2x$ are parabolas of the form $y = x^2 + C$

Visualizing Integral Curves



Explanation: The equation $\frac{dy}{dx} = 2x$ tells us that at each point (x, y), the slope of the curve is $2x$. All curves in the family $y = x^2 + C$ satisfy this condition.
Blue arrows show slope
At $x = -1.5$, slope $= 2(-1.5) = -3$
At $x = 1$, slope $= 2(1) = 2$

Example 6: Finding a Specific Integral Curve

Find the integral curve of $f(x) = 3x^2$ that passes through the point $(1, 4)$.

Solution:

1. General antiderivative: $\int 3x^2\,dx = x^3 + C$
2. So integral curves are: $y = x^3 + C$
3. Use the point $(1, 4)$: $4 = 1^3 + C \Rightarrow C = 3$
4. Specific curve: $y = x^3 + 3$

Verification:

- $\frac{dy}{dx} = 3x^2 = f(x)$
- When $x = 1$, $y = 1 + 3 = 4$

Differential Equations Viewpoint

Definition (Differential Equation)

An equation involving derivatives of a function is called a **differential equation**.

Simple Differential Equations

- $\frac{dy}{dx} = f(x)$ is a differential equation
- The solution is $y = \int f(x)\,dx = F(x) + C$
- An **initial condition** like $y(x_0) = y_0$ determines C

Example

Solve: $\frac{dy}{dx} = 2x$ with $y(0) = 1$
Solution: $y = \int 2x\,dx = x^2 + C$, and $1 = 0^2 + C \Rightarrow C = 1$, so $y = x^2 + 1$

Example 7: Solving Differential Equations

Solve the initial value problem: $\frac{dy}{dx} = \cos x$ with $y(0) = 2$

Solution:

1. General solution: $y = \int \cos x \, dx = \sin x + C$
2. Use initial condition: $y(0) = \sin 0 + C = 0 + C = 2 \Rightarrow C = 2$
3. Specific solution: $y = \sin x + 2$

Verification:

- $\frac{dy}{dx} = \cos x$
- $y(0) = \sin 0 + 2 = 2$

What are Slope Fields?

Definition (Slope Field)

A **slope field** (or direction field) is a graphical representation of a differential equation $\frac{dy}{dx} = f(x, y)$ that shows the slope of solution curves at various points.

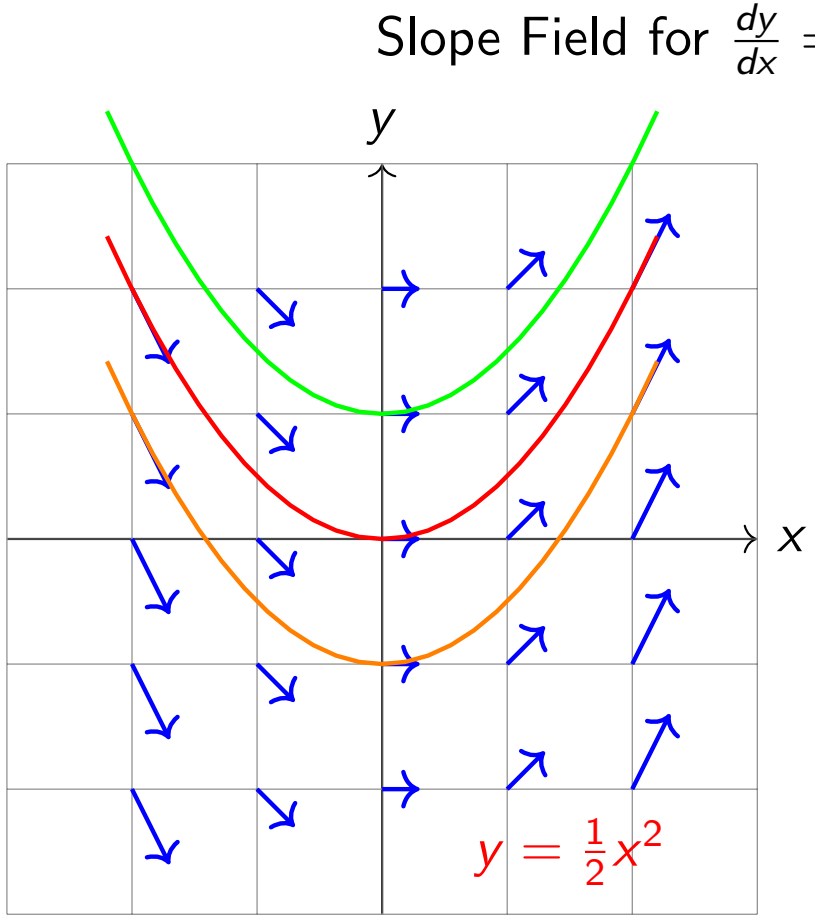
How to Construct Slope Fields

1. For $\frac{dy}{dx} = f(x)$, the slope depends only on x
2. At each x-value, draw small line segments with slope $f(x)$
3. Integral curves are curves tangent to these segments

Example

For $\frac{dy}{dx} = x$, at $x = -1$ slope is -1, at $x = 0$ slope is 0, at $x = 1$ slope is 1

Slope Field Visualization



The blue arrows show the slope at each point. The colored curves are solutions that follow these slopes.

Example 8: Using Slope Fields

Given the slope field for $\frac{dy}{dx} = -\frac{x}{y}$, sketch the solution curve through $(0, 2)$.

Observations:

- When $y > 0$, slope is negative for $x > 0$, positive for $x < 0$
- When $y < 0$, slope is positive for $x > 0$, negative for $x < 0$
- Slope is undefined when $y = 0$
- Solution through $(0, 2)$ should be a semicircle in upper half-plane

Solution: The curve is $x^2 + y^2 = 4$ with $y > 0$

Practice Problems - Basic Integration (8 problems)

Evaluate each indefinite integral:

1. $\int (5x^4 - 3x^2 + 2)\, dx$
2. $\int (2\sin x - 3\cos x)\, dx$
3. $\int \left(e^x + \frac{2}{x}\right)\, dx$
4. $\int (4x^3 + 2x - 1)\, dx$
5. $\int (\sec^2 x + \csc^2 x)\, dx$
6. $\int \left(3\sqrt{x} - \frac{1}{\sqrt{x}}\right)\, dx$
7. $\int (2e^x - 5\sin x + 3)\, dx$
8. $\int \left(\frac{1}{x^3} + 4x^2\right)\, dx$

Practice Problems - Applications (10 problems)

1. Find $f(x)$ if $f'(x) = 3x^2 - 2x$ and $f(1) = 4$
2. Find the integral curve of $f'(x) = \cos x$ through $(\pi, 2)$
3. Solve: $\frac{dy}{dx} = 4x^3$ with $y(0) = 1$
4. Find $f(x)$ if $f''(x) = 6x$ with $f'(0) = 1$ and $f(0) = 2$
5. A particle moves with velocity $v(t) = 3t^2 - 2t$. Find position $s(t)$ if $s(0) = 5$
6. The slope of a curve at (x, y) is $2x$. Find the equation if it passes through $(1, 3)$
7. Find the family of curves orthogonal to $y = x^2$
8. Solve: $\frac{dy}{dx} = e^x$ with $y(0) = 0$
9. Find the area function $A(x)$ if $A'(x) = 2x + 1$ and $A(0) = 3$
10. A population grows at rate $P'(t) = 100e^{0.1t}$. Find $P(t)$ if $P(0) = 1000$

Challenge Problems

1. Prove that all antiderivatives of $f(x) = 0$ are constant functions
2. Show that $\int [af(x) + bg(x)]\,dx = a\int f(x)\,dx + b\int g(x)\,dx$
3. Find a function f such that $f'(x) = f(x)$ for all x
4. Explain why $\int \frac{1}{x}\,dx = \ln|x| + C$ instead of $\ln x + C$
5. Create a slope field for $\frac{dy}{dx} = y$ and describe the solution curves

Summary of Key Concepts

- **Antiderivative:** $F'(x) = f(x)$
- **Indefinite Integral:** $\int f(x)\,dx = F(x) + C$
- **Power Rule:** $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1)$
- **Properties:** Linearity allows term-by-term integration
- **Integral Curves:** Family of antiderivative graphs
- **Differential Equations:** $\frac{dy}{dx} = f(x)$ solved by integration
- **Slope Fields:** Visual representation of differential equations

Remember

- Always include $+C$ for indefinite integrals
- Check your answer by differentiating
- Use initial conditions to find specific solutions
- Integration reverses differentiation